

Hussein Taiwo Lawal
Lagos, Nigeria | lawalhussein775@gmail.com | +234 903 501 4430
LinkedIn: <https://www.linkedin.com/in/lawal-hussein> | GitHub: <https://github.com/Tboiii-123>
Portfolio: <https://full-stack-developer-lemon.vercel.app/>

Profile

Python Developer with 3+ years of experience building and deploying high-performance web applications across fintech, SaaS, and logistics domains. Skilled in designing scalable RESTful APIs, optimizing relational databases, and implementing security best practices for production-ready systems. Proficient in authentication and authorization (JWT, OAuth2/OIDC), AI/ML integration, and third-party service integration (payment gateways, messaging APIs, analytics). Experienced in containerization with Docker and deployment on AWS, Render, and Heroku. Strong background in Agile development, TDD, and CI/CD, with a proven ability to deliver end-to-end solutions - from system architecture and database design to frontend integration, cloud deployment, and model inference pipelines.

Technical Skills

- **Programming Languages:** Python, JavaScript, Java
- **Frameworks & Libraries:** Django, Django REST Framework (DRF), FastAPI, Flask, React.js, Express.js
- **Frontend & UI:** HTML5, CSS3, Tailwind CSS, Bootstrap, Chart.js
- **Databases:** PostgreSQL, MongoDB, MySQL, SQLite, Supabase, Appwrite
- **Authentication & Authorization:** JWT, OAuth2/OIDC, Session-based Authentication
- **Payment Integrations:** Stripe, Flutterwave, Paystack
- **Cloud & DevOps Tools:** Docker, AWS, Heroku, Render, Git, Redis, Celery, Cloudinary, Zapier, RabbitMQ
- **Web Scraping & Automation:** Selenium, BeautifulSoup, Requests
- **Development Practices:** Test-Driven Development (TDD), CI/CD, REST API Design, Authentication & Security.

Professional Experience

Back End Developer | Dispatch Hub

- Developed and maintained a high-performance REST API for mobile app integration.
- Implemented JWT authentication for secure user access.
- Integrated Google Maps API for real-time location tracking.
- Utilized Redis for caching and fast data retrieval.
- Designed scalable endpoints with Django REST Framework.
- Ensured seamless interaction between mobile clients and the backend.
- Built robust, secure, and efficient APIs for real-time data access and improved user experience.

Key Skills & Technologies:

Django, Django REST Framework, Python, Redis, JWT Authentication, Google Maps API, API Development

Software Engineer – Altivox (Freelancer)

- Built and deployed a Logistics & Delivery Tracking System using Django, PostgreSQL, Tailwind, and Chart.js.
- Implemented session-based authentication for secure user and admin access.
- Developed an interactive KPI dashboard with Chart.js, enabling real-time performance insights.
- Designed advanced reporting and filtering features for shipment analysis and operational efficiency.
- Integrated async background tasks for email notifications and system alerts.
- Optimized database queries, boosting application performance by 35%.
- Enhanced delivery workflows, improving on-time delivery rates by 25% through live status tracking.

Python Developer – Bincom Dev Center, Lagos

- Designed and implemented Django-based applications with secure authentication and REST APIs.
- Applied data science libraries (Pandas, NumPy, Scikit-learn) to prototype analytics and predictive features.

Software Engineer – Power Learn Project

- Selected as part of the 2024 PLP Africa Cohort 4, a competitive training program for emerging software engineers.
- Expanded expertise across web development (HTML, CSS, JavaScript) and mobile app development with Dart and Flutter.
- Strengthened backend development skills through hands-on projects using Django and Python programming.
- Collaborated with peers in an Agile learning environment to deliver functional, real-world applications.

Projects

E-commerce Web Application (Python + Django)

- Built a full-stack e-commerce platform enabling users to browse, add to cart, and securely checkout using integrated payment gateways (Paystack/Stripe).
- Implemented email-based authentication, password reset, and role-based access control (admin/customers).
- Designed normalized PostgreSQL schema, achieving 30% faster query response times through indexing and optimization.
- Integrated Celery with Redis for order notifications and automated email receipts.
- Deployed to Render with environment-specific configurations and managed product images via Cloudinary.

Vending Machine Platform (Python + Django + DRF + Swagger docs + Pytest + JWT Authentication)

- Developed a full-featured vending machine API with Django REST Framework, supporting product management, purchases, deposits, and session handling.
- Implemented role-based access control: ****buyers**** can deposit coins, buy products, and reset deposits; ****sellers**** can create, update, and delete products.
- Designed robust purchase logic: verifies product availability, calculates total cost, manages user deposits, and returns optimal change using a custom ``calc_change`` utility.
- Built JWT-based authentication with active session tracking, preventing concurrent logins and enabling secure session termination.
- Integrated Swagger documentation (``drf_yasg``) for all endpoints, including request/response schemas and automatic API exploration.
- Applied comprehensive testing using ``pytest`` with Django test client, achieving ****100% coverage**** for views, serializers, and permission rules.
- Designed serializers with validation rules (e.g., product cost must be multiples of 5) to enforce business constraints and improve data integrity.
- Structured a maintainable, modular backend with clear separation of concerns for serializers, views, utils, and models.

-News Article Microservice *(Node.js, Express, PostgreSQL, Multer, JWT Authentication, BunnyCDN+ Swagger + Redis)*

- Built a scalable microservice for managing news articles, integrated with the platform's authentication system for secure user interactions.
- Implemented PostgreSQL-based relational data storage with caching and pagination for efficient performance.
- Added notifications, logging, and robust error handling to enhance user experience and maintainability.
- Managed image uploads and storage using Multer and BunnyCDN.
- Created comprehensive API documentation with Swagger for seamless frontend and React Native integration.
- Secured endpoints with JWT authentication and implemented full CRUD functionality, including reporting features for news articles.

Referral Email Automation Platform (Python + Django)

- Re-engineered the legacy email notification system by replacing direct API calls with a Zapier-based automation workflow.
- Implemented a Django API (``start_workflow``) that receives JSON payloads, validates input, and securely forwards data to Zapier webhooks.
- Added environment-aware logic (staging vs production) for safe testing and reliability.
- Designed exception handling with custom logging to improve resilience and error tracking.
- Built a React.js frontend (App Router) with a form-based UI to trigger referral workflows seamlessly.
- Centralized all third-party email operations into a maintainable automated pipeline, reducing integration overhead.

Multi-Tenant Project Management API (FastAPI + Swagger doc + JWT Authentication)

- Designed and implemented a FastAPI backend to manage tenants, users, and projects with role-based access control.
- Built authentication and authorization using JWT, including login, profile endpoints, and admin-only permissions.
- Implemented project CRUD tied to tenant ownership, ensuring isolation between companies.
- Created separate flows for user signup and tenant admin creation for company-level onboarding.
- Added validation and exception handling for secure, predictable API responses.
- Integrated SQLAlchemy ORM with PostgreSQL (or SQLite for dev) and documented endpoints with Swagger UI.

Video API (Express.js + TypeScript + Swagger doc + JWT Authentication)

- Developed a RESTful API for managing videos with authentication using Express.js and TypeScript.
- Implemented JWT-based authentication for secure user registration and login.
- Built full CRUD functionality for videos, including filtering by genre and pagination.
- Integrated Swagger for automatic API documentation.
- Utilized TypeORM with SQLite to manage data persistence.
- Wrote unit and integration tests with Jest and Supertest to ensure reliability.

Education

Aptech, Lagos – Advanced Diploma in Software Engineering

NIIT, Lagos – Python Programming Certificate